

Design a 4:1 Rover Gearbox

Design and justify a gear train that provides an approximately 4:1 speed reduction for a rover drive system.

Intermediate	2 class periods	Engineering Challenge
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Engineering Context

A rover motor spins too quickly and produces insufficient wheel torque. Your mechanical team must create a compact reduction system using available gears.

What You Will Practice

- Calculate gear ratio
- Relate speed and torque
- Compare multiple gear-train layouts
- Document a mechanical recommendation

What You Need

- Available classroom gears or VEX gear inventory
- Axles and supports or CAD layout tools
- Calculator
- Engineering notebook

Student Instructions

- 1 Inventory available gear tooth counts.
- 2 Create three possible gear-train combinations near a 4:1 reduction.
- 3 Calculate theoretical ratio and output speed for each.
- 4 Sketch or model each layout and identify required shafts and supports.
- 5 Compare size, number of parts, direction of rotation, and likely friction.
- 6 Select one design and build or mock up the gear train.
- 7 Measure input and output rotations to verify the actual ratio.
- 8 Recommend the best design for the rover.

Engineering Requirements

- Target reduction between 3.8:1 and 4.2:1
- Output shaft must rotate continuously without gear interference
- All calculations must show driver and driven gears
- Recommendation must compare at least three concepts

Constraints & Safety

- Use only available gears and approved structural parts
- Do not apply motor power until alignment is checked by hand
- Keep hair and fingers clear of moving gears

What You Submit

- Three concept layouts
- Gear-ratio calculations
- Built or CAD-modeled final design
- Measured ratio data
- Recommendation memo

Reflection Questions

- 1 Why might actual performance differ from the theoretical ratio?
- 2 Which layout used the fewest parts?
- 3 How did the reduction affect output torque and speed?
- 4 What support would prevent gear skipping?

Engineering Tip For compound gear trains, multiply the ratio of each stage rather than adding ratios.	Common Mistake Using pitch diameter or tooth count inconsistently will produce an incorrect ratio.
Stretch Goal Redesign the gearbox to provide two selectable ratios.	Career Connection Mechanical and drivetrain engineers size gear reductions for robotic explorers, aircraft actuators, and test equipment.